

Evaluating MATHstream

IRAISE Workshop, Festival of Learning 2026
Seoul, Republic of Korea

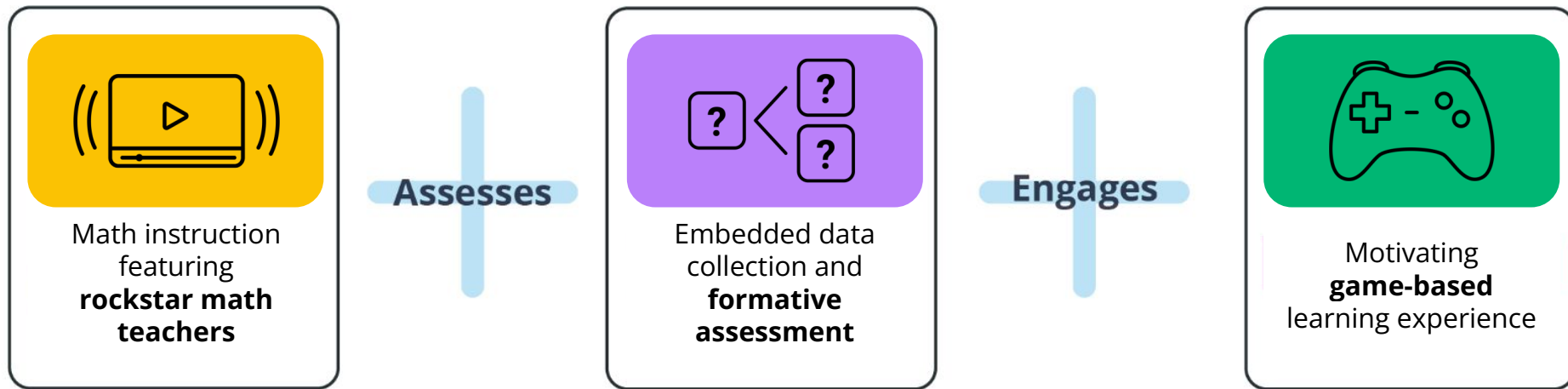
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Agenda

- What is MATHstream?
 - What is LiveHint AI?
- Preliminary evidence of associations with learning outcomes
- SCALE Lab
- Evaluating MATHstream in a large-scale RCT
 - Planning & challenges

MATHstream is...

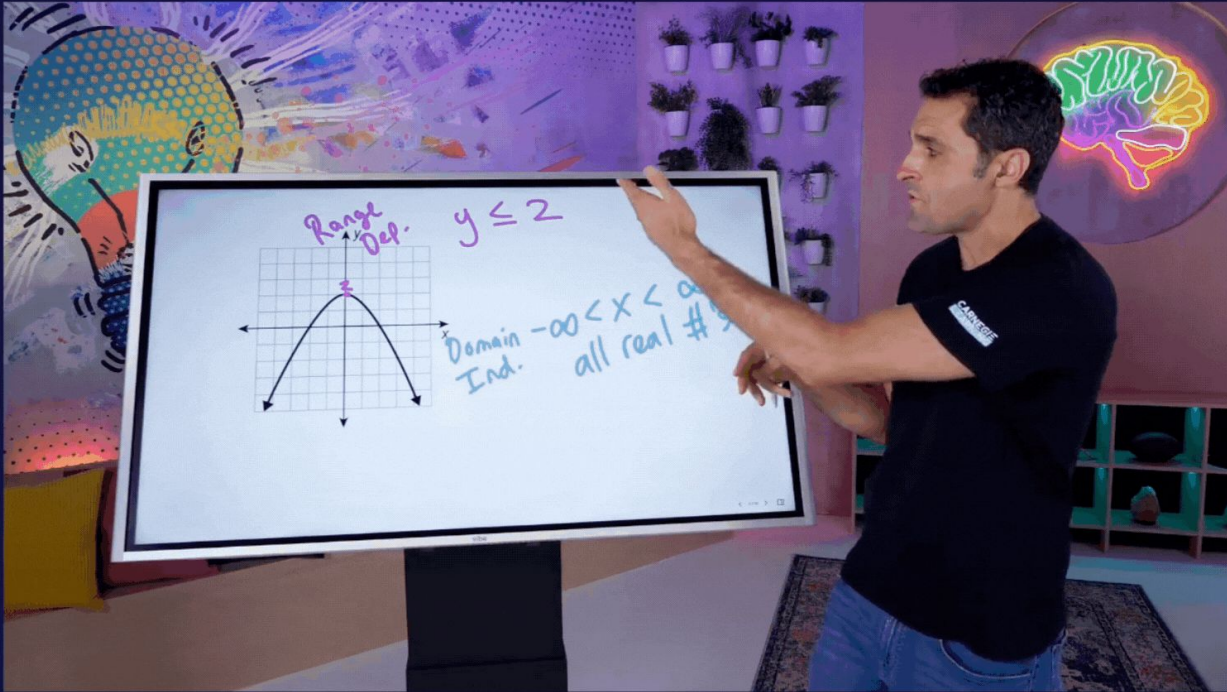


generally used as a supplement to Carnegie Learning's core, blended math curriculum.

MATHstream in action

MATHstream

Tutorbot hints



The whiteboard displays a graph of a downward-opening parabola on a coordinate plane. The vertex is at (0, 2). The graph is labeled with "Range" and "Def." in purple, and the inequality $y \leq 2$ is written in purple. The domain is labeled "Domain" and "Ind." in blue, with the inequality $-\infty < x < \infty$ and the text "all real #s" written in blue.

Video player controls: Play/Pause, Progress bar, Closed Captions (CC), Audio, Full Screen.

LiveHint AI in MATHstream

- Socratic-style multimodal AI tutoring system
 - text, audio, avatar-based video, and real-time graph generation.
- Curriculum specific with item-specific system instructions.
- Guardrails for toxic, off-track text, and bias.
 - Evaluated for explicit and implicit demographic identifiers with different underlying LLMs*
- Integrated with UpGrade for A/B testing.

*A. Vinodh, E. Harvey, H. Almoubayyed, R. Yu, C. Brooks, A. Koenecke, R. Kizilcec (2025). Evaluating an AI Tutor for Bias Across Different Foundation Models, *AIED 2025*.

LessonStream: Ordering Part-to-Part and Part-to-Whole Ratios : 30 coins

> Type the answers in each space provided. Use numbers instead of words.

A soccer team was awarded a penalty shot at the end of a tie game. If the team makes the penalty shot, it will win the league championship. The coach is considering three players to take the penalty.

• Amber has taken 4 penalty shots this season and has made 3 of them.

• Lindsay has taken 6 penalty shots and made 4.

• Li has taken 3 penalty shots and made 2.

Determine the ratio of shots made to shots taken for each player. Write each ratio as a fraction.

Answer

Amber:

Lindsay:

Li:

> Select the correct answer from the drop-down menu.

Which player would you recommend take the penalty shot?

should take the penalty shot because they have the greatest ratio of shots made to shots attempted.

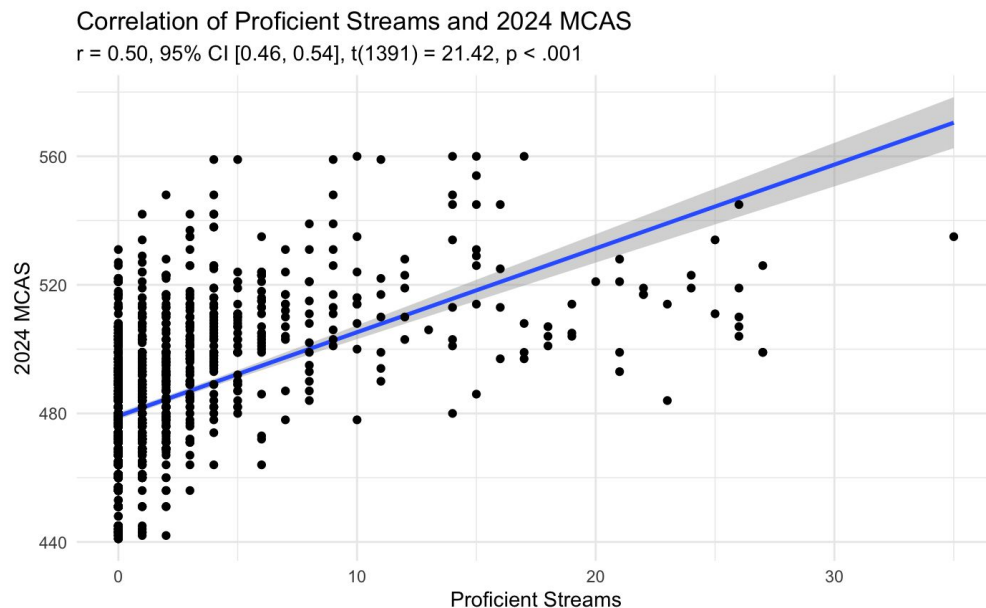
The screenshot shows the LiveHint AI interface. At the top, there's a header with the LiveHint AI logo (a star and a plus sign), the word "BETA", and navigation buttons for "Video", "Audio", and "Text". Below the header, the main content area displays a math problem about penalty shots and ratios. To the right of the problem is a chat window with a dark background. The chat window contains several messages: a welcome message, a disclaimer about AI interactions, a message about the AI's capabilities, and a "Let's Go!" button. Below these are three example prompts: "Let's start by identifying the final goal of this problem. Is it to:", "Recommend the player who should take the penalty by comparing their shots-made to shots-taken fractions.", and "That's correct! Let's start by determining the ratio of shots made to shots taken for each player. What are the numbers of shots made and taken for Amber, and which number goes on top of the fraction?". At the bottom of the chat window, there's a "Say goodbye to end this chat." button and a text input field with a "Type a message..." placeholder and a send button.

MATHstream & better EoY test performance

Correlational Analysis (n = 1,039 in MA, USA):

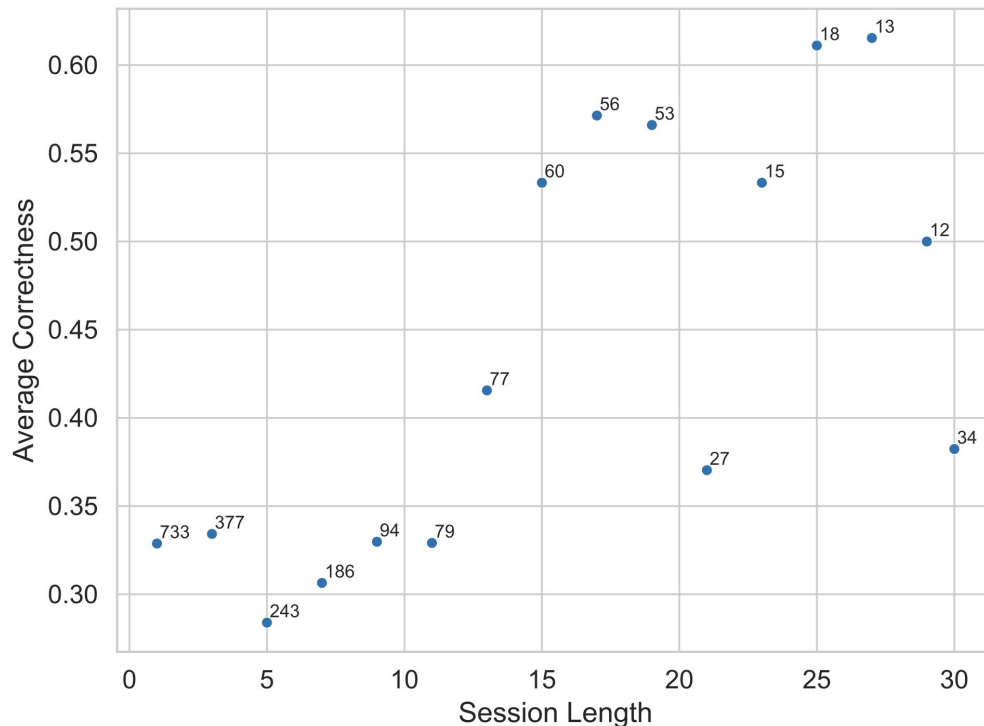
- Multilevel linear models with robust covariates (prior & current year math achievement, demographics, etc.)
- Proficient streams in MATHstream vs. EoY math test outcomes (MCAS = MA State Test)
- Each proficient stream is associated with **+2.61 MCAS points on average**.
- Adjusting for demographics, prior year score, and MATHia performance: **+0.65 MCAS points**.

Matched analyses find students with greater MATHstream usage significantly out-perform lower & no usage students by **+5-6 MCAS** points.



How is LiveHint AI Usage Associated with MATHstream Performance?

- Students who have longer sessions with LiveHint AI are more likely to get the problem right.
- 998 students, 2087 sessions, controlling for school and problem-level effects: correctness odds higher by 46% for each 10 turns.



H. Almoubayyed, Z. Lee, T. Cowell, K. Norberg, L. De Ley, U. Shaibu, A. Murphy, S.E. Fancsali, and S. Ritter. *Manuscript in prep.* **[PLEASE DO NOT CITE THIS SLIDE'S CONTENT WITHOUT PERMISSION]**

Enter RCT evaluation with SCALE Initiative

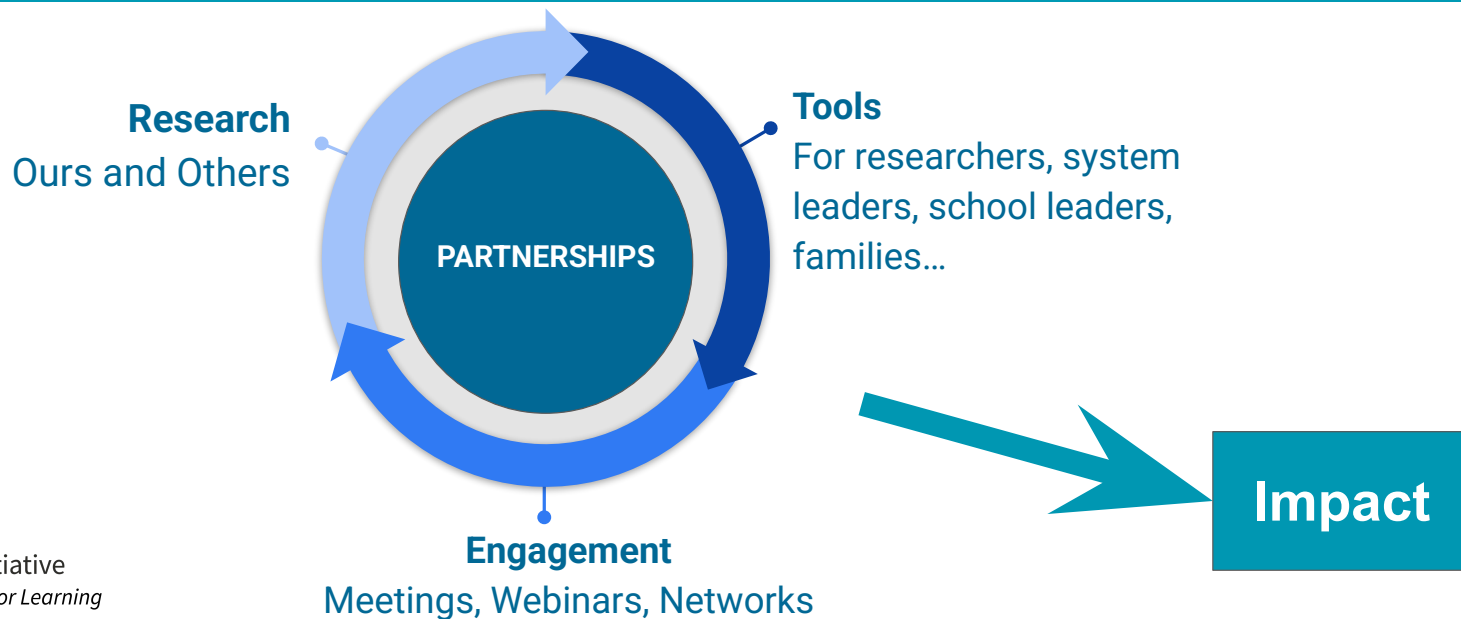
- Preliminary evidence of statistically significant associations between MATHstream usage/performance & end-of-year math assessment outcomes with high-quality matching & co-variates.
 - Even relatively low usage of MATHstream associated with improved outcomes: could suggest importance of teacher practices/implementation?
- Additionally, preliminary evidence of associations between usage of LiveHint AI and performance in MATHstream.

→ Now seek to establish more rigorous evidence of efficacy/effectiveness via a randomized controlled trial.

SCALE Initiative & Evaluating MATHstream

Help to Scale the Most Promising Practices Through...

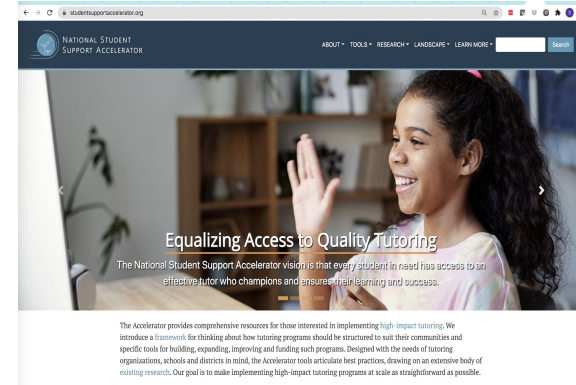
Motivation: A vast range of decision makers affect students' access to educational opportunities, so that even the most promising approaches may not scale to reach students who could benefit without effort to understand the needs and opportunities across contexts and to engage stakeholders



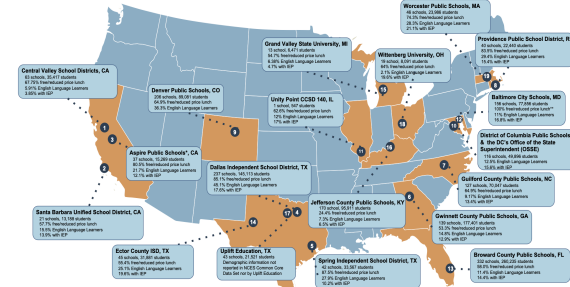
National Student Support Accelerator

- Unusually strong research base
- 20+ fast turnaround studies
 - o such as virtual, short-burst, opt-in...
- Materials for federal and state government and districts
- Expanded to 45% of schools even after pandemic funding

nssa.stanford.edu



National Student Support Accelerator Pilot Sites



These locations were the locations of the first 100+ pilot sites. Additional pilot sites are being added. Pilot sites are not limited to the locations shown on this map. Pilot sites are not limited to the locations shown on this map. Pilot sites are not limited to the locations shown on this map.

Now Use Approach for Scaling Promising AI Tools for Education

- **But...**
 - o The focus is much broader and less bounded
 - o The research base is much weaker
 - o The tools and programs are changing much more rapidly

need evidence to steer towards benefit while keeping up with decision-making timing

Carnegie Learning and the MATHstream Intervention

“Maximum support model”

We expect the impact of MATHstream to be the greatest as:

1. Support for students of **“developing” teachers***.
2. Support for **absent students and teachers**.
3. **Tier II** Intervention.

Absenteeism and prevalence of developing teachers are particularly high in districts serving low-income students.

**Working definition of developing teachers: Long-term subs, uncertified and emergency certified math teachers, and certified teachers with three or fewer years of math teaching experience.*

Large-scale RCT Design

Recruitment:

- Isolate effect of **MATHstream**
 - ◆ Supplemental tool to Carnegie Learning Core blended math
- Recruit from CL Core blended math districts

Intervention

Control group: Business as usual - CL Core blended math solution.

Treatment group: CL Core blended math solution + MATHstream access.

Randomization Unit:

- Combination of teacher and grade-level
 - ◆ Teacher practice: teacher-level gives most statistical power
 - ◆ However, some districts restrict to grade-level
- Stratification by school: all schools get something.

Measuring Impact

Outcomes:

- Student Math Assessment Scores
- Implementation:
 - ◆ Attendance
 - ◆ MATHstream assignment and usage
 - ◆ Platform engagement and performance metrics

Heterogeneity:

- Teacher and Student Prior Absences

Thank you

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